MODULE 11 · PRACTICE WORKSHEET

Fundamentals of Strings & Objects

Task 1.1 — Spot the Difference

```
let str = "hello";
let arr = ["h", "e", "l", "l", "o"];
```

- Log str.length and arr.length. What do you notice?
- Try str[0] and arr[0]. Do both work the same way?
- Try str.push("!") — what happens? Why does this fail for strings but not arrays?

Task 1.2 — Convert Between Them

- Convert the string "JavaScript" into an array of characters.
- Convert the array ["c", "a", "t"] back into a single string "cat".

Task 2.1 — Case Conversion

```
let name = "JavaScript";
```

- Print the uppercase version.
- Print the lowercase version.
- Print the string with whitespace trimmed: "   hi there   " → "hi there"

Task 3.1 — Slicing Strings

```
let sentence = "Learning JavaScript is fun!";
```

- Get just the word "Learning" using .slice().
- Get the last 4 characters ("fun!") using .slice() with a negative index.

Task 3.2 — Combine Strings

- Use .concat() to join "Hello" and "World" into "Hello World".
- Use + and template literals to do the same thing. Compare all three approaches.

Task 4.1 — Three Reversal Methods

Write a function reverseString(str) three different ways:

- Using .split(""), .reverse(), and .join("")
- Using a for loop that builds the reversed string character by character
-

Test all three with "JavaScript" → should return "tpircSavaJ".

Bonus: Which method do you think is fastest? Why?

Task 5.1 — Build a Profile Object

Create an object student with these properties: name, age, grade, and isEnrolled.

- Log the whole object.
- Log just the name property using dot notation.

Task 5.2 — Watch the Video Concept

After watching the "What is an Object" video, write 2–3 sentences (as a comment in your code) explaining an object in your own words, using a real-life analogy (not a car or a person — get creative!).

Task 6.1 — Dot vs Bracket Notation

```
let car = { brand: "Toyota", model: "Corolla", year: 2022 };
```

- Get brand using dot notation.
- Get model using bracket notation.
- Add a new property color: "blue" using bracket notation.
- Update year to 2023 using dot notation.

Task 7.1 — Keys and Values

```
let book = { title: "The Hobbit", author: "Tolkien", pages: 310 };
```

- Get an array of all keys
- Get an array of all
- Delete the pages property using delete.

Task 7.2 — Nested Objects

```
let user = {
  username: "coder123",
  address: {
    city: "Austin",
    zip: "78701"
  }
}
```

- Log the city using dot notation chaining.
- Add a country property inside the address.
- Delete the zip property from the nested object.

Task 8.1 — Loop Through Properties

```
let scores = { math: 90, science: 85, art: 95 };
```

- Use a for...in loop to log each key and value like: math: 90.
- Calculate the average of all values in the loop.

Task 9.1 — Contact Book

Build a small "contact" system:

```
let contact = {
  name: "Alex Johnson",
  email: "ALEX@EMAIL.COM",
  phone: "555-1234"
};
```

1. Convert the email to lowercase and update the object.
2. Loop through the contact object and print each key-value pair.
3. Add a new property favoriteWords: [] (an array) — push 3 words to it.
4. Reverse the name string just for fun using one of your three reverse functions from Task 4.1.
5. Check if the email includes "@email.com" (case-insensitive) and log a confirmation message.
